KOSZULNESS CHART TORSORS AND CHIRAL KOSZULNESS CRITERIA

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Abstract. For a chiral algebra $\mathcal{A}$ on a smooth projective curve X with PBW filtration, the Koszulness locus is a derived formal moduli problem over the Drinfeld associator torsor. At finite PBW stage and finite GRT_1 weight quotient, after fixing an associator coordinate and a gauge framing, its classical truncation is an open subscheme of a finite-type affine Maurer–Cartan scheme. Unframed finite stages are quotient stacks; the full completed object is a derived inverse limit. Nonempty framed components carry the resulting pro-GRT_1 chart torsor.

The intrinsic Koszulness core consists of the chiral Koszul twisting datum, PBW E_2 collapse, A_∞ formality of the bar model, and bar–cobar counit inversion in the Theorem-B ambient. Ext diagonal concentration is a consequence, with a converse under the chiral BGS/Priddy diagonal-detection hypothesis. Kac–Shapovalov nondegeneracy is a family criterion. Barr–Beck–Lurie monadicity is a consequence of the counit. Factorization homology is a criterion only under the stated comparison and detection hypotheses. Chiral Hochschild concentration is a one-way consequence on the Koszul locus; a free-polynomial cup-product description requires Massey vanishing. The Lagrangian criterion is conditional on perfectness and nondegeneracy, and $\mathcal{D}$-module purity is used in the proved direction toward boundary acyclicity.

1. Introduction

1.1. The problem. The Koszul duality of a graded associative algebra $A = T(V)/(R)$ with quadratic relations $R \subseteq V \otimes V$ admits many equivalent formulations. Priddy's original criterion is purity of the bar complex [13]. Beilinson, Ginzburg, and Soergel extend this to the derived category of a highest-weight category and give a cohomological characterization via Ext diagonal vanishing [2]. Loday and Vallette place Koszul duality in the operadic framework and reformulate it as an A_∞-formality condition for the Koszul dual cooperad [11]. Polishchuk and Positselski develop the dual story for corings and comodules [12]. Each of these reformulations, in the associative world, has a parallel in the world of chiral algebras (equivalently, vertex algebras) over an algebraic curve: the bar complex carries an extra layer of factorization geometry, the Koszul dual inherits an $\mathcal{O}$-module structure, and the relevant spectral sequence lives on a Fulton–MacPherson compactification.

The classical web of Koszul criteria survives in the chiral world only after its scope is separated. Seven criteria form the intrinsic bidirectional core. Kac–Shapovalov nondegeneracy and Fulton–MacPherson boundary acyclicity are chiral criteria inside that core. Barr–Beck–Lurie monadicity is a consequence of the bar–cobar counit. Factorization homology requires comparison and detection hypotheses. Chiral Hochschild concentration is a one-way theorem on the Koszul locus. The Lagrangian and $\mathcal{D}$-module tests retain their own perfectness and purity hypotheses.

1.2. Koszulness moduli. The Koszulness moduli object is controlled by the completed convolution dg Lie algebra of the bar–cobar counit. It is a derived formal moduli problem before truncation. A scheme appears only after finite PBW truncation, finite GRT_1 weight quotient, associator coordinate, and gauge framing.

Definition 1.1 (Koszulness moduli object)**.** For a complete local Artinian commutative dg $\mathbb{Q}$-algebra R with maximal ideal $\mathfrak{m}_R$, set

$$\widehat{\mathfrak{g}}_\Phi(\mathcal{A}) = \varprojlim_N \mathrm{Conv}_\Phi\big(F_{\le N}\bar{B}^{\mathrm{ch}}(\mathcal{A}), F_{\le N}\mathcal{A}\big).$$

The Koszulness moduli object is the derived functor

$$\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})\colon \mathbf{Art}^{\wedge}_{\mathbb{Q}} \longrightarrow \mathbf{Spaces},$$

$$R \longmapsto \mathrm{hofib}_0\left[\coprod_{\Phi\in\mathrm{Assoc}(R)} \mathrm{MC}_\bullet\big(\widehat{\mathfrak{g}}_\Phi(\mathcal{A})\widehat{\otimes}\mathfrak{m}_R\big) \xrightarrow{\xi\mapsto\mathrm{Cone}(\varepsilon_\xi)} D^{\mathrm{co}}_{\mathrm{ch}}(X;R)\right],$$

where $\varepsilon_\xi\colon \Omega^{\mathrm{ch}}_\Phi(\widehat{\bar{B}^{\mathrm{ch}}}(\mathcal{A}),\xi) \to \mathcal{A}\widehat{\otimes}R$ is the deformed counit. An R-point is a triple (Φ,ξ,h_ξ) consisting of an associator, a Maurer–Cartan element, and a trivialisation of $\mathrm{Cone}(\varepsilon_\xi)$ in the coderived chiral category over R.

Theorem 1.2 (Koszulness moduli, finite-stage representability)**.** *Let $\mathcal{A}$ be of finite PBW type. At each finite PBW stage and finite weight quotient of* GRT_1*, assume PBW–Artin type: the finite convolution dg Lie algebra is nilpotent with finite-dimensional bounded cohomology, the gauge group is unipotent algebraic, the finite Koszul cone is perfect with coherent cohomology, and the tower is derived Mittag–Leffler. Then* $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$ *is a possibly empty derived formal moduli problem over the Drinfeld associator torsor. The unframed finite-stage truncation is a derived Artin quotient stack. After fixing an associator coordinate and a gauge framing, its classical truncation is an open subscheme of a finite-type affine Maurer–Cartan scheme, cut out by acyclicity of the finite Koszul cone. The completed object is the derived inverse limit of these finite stages.*

The moduli object is nonempty if and only if the completed bar–cobar counit $\Omega^{\mathrm{ch}}(\widehat{\bar{B}^{\mathrm{ch}}}(\mathcal{A})) \to \mathcal{A}$ *is a quasi-isomorphism. On each nonempty framed component, the projection to the associator torsor is a pro-*GRT_1 *torsor. Before framing the same component is the quotient of that torsor by the automorphism group of the witness.*

1.3. **Classical Koszulness criteria.** Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve X, equipped with its Poincaré–Birkhoff–Witt filtration $F_\bullet\mathcal{A}$.

Theorem 1.3 (Equivalences and consequences of chiral Koszulness)**.** *The following intrinsic conditions are equivalent:*

- (i) $\mathcal{A}$ *is chirally Koszul in the sense of Definition 2.3;*
- (ii) *the PBW spectral sequence on* $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ *collapses at* E_2*;*
- (iii) *the minimal* A_∞*-model of* $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ *is formal:* $m_n = 0$ *for* $n \ge 3$*;*
- (iv) $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X,\omega_X) = 0$ *for* $p \ne q$*;*
- (v) *in the Theorem-B ambient, with* $C_{\mathcal{A}} = \bar{B}^{\mathrm{ch}}(\mathcal{A})$ *on raw finite PBW windows and* $C_{\mathcal{A}} = \widehat{\bar{B}^{\mathrm{ch}}}(\mathcal{A})$ *on the strict completed pro-conilpotent regime, the continuous bar–cobar counit* $\Omega^{\mathrm{cont}}_X(C_{\mathcal{A}}) \to \mathcal{A}$ *is a quasi-isomorphism;*
- (ix) *the Kac–Shapovalov determinant* $\det G_h \ne 0$ *in the bar-relevant range;*
- (x) *for every Fulton–MacPherson boundary stratum S, the restriction* $i_S^!\,\bar{B}_n(\mathcal{A})$ *is acyclic outside degree* 0*.*

The Barr–Beck–Lurie comparison functor for the bar–cobar adjunction is a consequence of the counit criterion. Factorization homology concentration is a criterion only under the comparison and detection hypotheses stated in Theorem 3.1. On the Koszul locus, $\mathrm{ChirHoch}^*(\mathcal{A})$ *has cohomological amplitude* $[0,2]$

and Hochschild duality; a free-polynomial cup-product description is conditional on Massey vanishing. Under a perfectness and non-degeneracy hypothesis on the ambient tangent complex, condition

(xi) *the Koszul locus subfunctor $\mathcal{M}_{\mathcal{A}}$ and its dual $\mathcal{M}_{\mathcal{A}^!}$ are transverse Lagrangians in the* (-1)*-shifted symplectic deformation space* $\mathcal{M}_{\mathrm{comp}}$

is equivalent to them. Finally,

(xii) *each $\bar{B}^{\mathrm{ch}}_n(\mathcal{A})$ is pure of weight n as a mixed Hodge module, with characteristic variety aligned to the FM boundary strata*

implies (x)*; the converse is proved for the affine Kac–Moody family, and open in general.*

The affine Kac–Moody family is controlled by the Kac–Shapovalov determinant. The $\beta\gamma$ system is controlled by the free PBW bar model. The Virasoro Koszul locus is controlled by PBW collapse. These three tests land in the same intrinsic core.

1.4. **History and related work.** Priddy [13] introduced the Koszul property for quadratic algebras via resolutions. Beilinson, Ginzburg, and Soergel [2] showed that Koszulness is equivalent to Ext diagonal vanishing in a broad categorical context, a formulation that survives in (iv). Polishchuk and Positselski [12] carried the theory to noncommutative, graded, and curved settings. Loday and Vallette [11] organized the operadic framework and characterized Koszulness as A_∞-formality of the Koszul dual cooperad, matching condition (iii). Positselski's construction of the Koszul duality functors between modules and comodules is the associative prototype of the chiral bar–cobar functors used below.

On the chiral side, Beilinson and Drinfeld [1] developed factorization algebras and chiral algebras on a curve and isolated the role of the PBW filtration. Francis and Gaitsgory [4] gave an adjunction between chiral Lie and chiral commutative operads on $\mathrm{Ran}(X)$, which is the operadic backbone of the bar–cobar adjunction here. Gaitsgory and Rozenblyum [5] give the ∞-categorical framework adequate to formulate condition (vi). Costello and Gwilliam [3] build the factorization homology machinery used in (vii). On the Kac–Shapovalov side, the vanishing of the Shapovalov determinant in specific ranges is a classical criterion for irreducibility [7]; its role in controlling the PBW spectral sequence was hinted at in the work of Feigin–Fuchs on Virasoro representations.

The synthesis is the derived chart atlas $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$ together with the intrinsic equivalence core of Theorem 3.1.

1.5. **Conventions.** We work in characteristic zero. All chain complexes are cohomologically graded, $|d| = +1$. Desuspension lowers degree: $|s^{-1}v| = |v| - 1$. The bar complex uses the augmentation ideal $\bar{\mathcal{A}} = \ker(\varepsilon)$, not $\mathcal{A}$ itself; we write $\bar{B}^{\mathrm{ch}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ for the geometric chiral bar complex, with deconcatenation coproduct. When we refer to the symmetric bar $\bar{B}^{\Sigma}(\mathcal{A})$, we specify it explicitly. Imported results from the monograph are cited at the point where their hypotheses enter.

2. Preliminaries

2.1. **Chiral algebras and the PBW filtration.** Let X be a smooth projective curve over an algebraically closed field of characteristic zero. A *chiral algebra* on X is a left $\mathcal{D}_X$-module $\mathcal{A}$ equipped with a chiral bracket

$$\mu\colon\ j_*j^*(\mathcal{A}\boxtimes\mathcal{A}) \longrightarrow \Delta_*\mathcal{A},$$

where $j: X^2\setminus\Delta \hookrightarrow X^2$ is the complement of the diagonal and $\Delta\colon X \to X^2$ is the diagonal embedding. The bracket is required to satisfy the Beilinson–Drinfeld axioms of skew-symmetry and Jacobi, equivalent in the vertex algebra language to the Borcherds identity [1].

We write $\varepsilon\colon \mathcal{A} \to \omega_X$ for the augmentation (the projection to the vacuum), and $\bar{\mathcal{A}} = \ker(\varepsilon)$ for the augmentation ideal. The *PBW filtration* $F_\bullet\mathcal{A}$ is the increasing filtration generated by placing the generators in $F_1\mathcal{A}$ and closing under the chiral bracket. We write $R_{\mathcal{A}} = \mathrm{gr}^F_\bullet \mathcal{A}$ for the associated graded; it is a Poisson vertex algebra [9]. The filtration is bounded below (by $F_0 = \omega_X$) and exhaustive.

2.2. **The geometric bar complex.** Fix $n \geq 1$. The bar complex in degree n is constructed on the Fulton–MacPherson compactification $\overline{C}_n(X)$ of the configuration space $C_n(X) = X^n \setminus \bigcup_{i<j} \Delta_{ij}$. As a graded $\mathcal{D}_{\overline{C}_n(X)}$-module,

$$\bar{B}^{\mathrm{ch}}_n(\mathcal{A}) \;=\; j_* j^* \bar{\mathcal{A}}^{\boxtimes n} \otimes \Omega^{n-1}_{\overline{C}_n(X)}(\log D)[1-n],$$

where D is the boundary divisor and the shift $[1-n]$ is the desuspension grading $|s^{-1}v| = |v| - 1$ applied $n-1$ times. The assembly

$$\bar{B}^{\mathrm{ch}}(\mathcal{A}) \;=\; \bigoplus_{n\geq 1} \bar{B}^{\mathrm{ch}}_n(\mathcal{A})$$

carries the deconcatenation coproduct and a differential $d_{\mathrm{bar}} = d_{\mathcal{A}} + d_{\mathrm{OPE}} + d_{\mathrm{dR}}$ whose three components encode the internal $\mathcal{D}$-differential, the OPE residue along the collision divisors, and the de Rham differential on $\overline{C}_n(X)$ [1, 10].

Remark 2.1. The use of the augmentation ideal $\bar{\mathcal{A}}$ is essential: writing $T^c(s^{-1}\mathcal{A})$ in place of $T^c(s^{-1}\bar{\mathcal{A}})$ would include the vacuum and destroy the bar-cobar adjunction because the vacuum summand is not part of the reduced bar coalgebra.

2.3. **The cobar functor.** Dually, for a factorization coalgebra C on X with coaugmentation $\eta\colon \omega_X \to C$, the cobar complex is

$$\Omega^{\mathrm{ch}}(C) \;=\; \bigoplus_{n\geq 1} (s\bar{C})^{\otimes n} \otimes \Omega^{n-1}_{\overline{C}_n(X)}(\log D)[n-1],$$

where $\bar{C} = \mathrm{coker}(\eta)$ and the differential is the dual of the bar differential.

Proposition 2.2 (Bar–cobar adjunction; [10]). *The functors $\bar{B}^{\mathrm{ch}} \dashv \Omega^{\mathrm{ch}}$ form an adjoint pair between chiral algebras and factorization coalgebras on X. The unit $\eta_C\colon C \to \bar{B}^{\mathrm{ch}}(\Omega^{\mathrm{ch}}(C))$ and counit $\varepsilon_{\mathcal{A}}\colon \Omega^{\mathrm{ch}}(\bar{B}^{\mathrm{ch}}(\mathcal{A})) \to \mathcal{A}$ are natural transformations.*

2.4. **Chiral twisting morphisms and Koszulness.** Let $\tau\colon C \to \mathcal{A}$ be a twisting morphism: a degree $+1$ map of $\mathcal{D}_X$-modules satisfying the Maurer–Cartan equation $d\tau + \tau \star \tau = 0$ with respect to the convolution product. The *left twisted tensor product* $K^L_\tau(\mathcal{A}, C)$ is the complex $\mathcal{A} \otimes C$ with differential $d_{\mathcal{A}} \otimes 1 + 1 \otimes d_C + (\mu \otimes 1)(1 \otimes \tau \otimes 1)(1 \otimes \Delta_C)$; the right version $K^R_\tau(C, \mathcal{A})$ is defined symmetrically.

Definition 2.3 (Chiral Koszul morphism). A chiral twisting datum $(\mathcal{A}, C, \tau, F_\bullet)$ is *Koszul* if:

(a) $\mathrm{gr}_F\, \mathcal{A}$ is classically Koszul on each finite PBW window;
(b) both $K^L_\tau(\mathcal{A}, C)$ and $K^R_\tau(C, \mathcal{A})$ are acyclic;
(c) the PBW filtration is complete, separated, and Mittag–Leffler on finite windows, so associated-graded acyclicity lifts to the completed chiral bar complex.

The chiral algebra $\mathcal{A}$ is *chirally Koszul* if there exists a Koszul twisting datum of the form $(\mathcal{A}, \bar{B}^{\mathrm{ch}}(\mathcal{A}), \tau_0, F^{\mathrm{PBW}}_\bullet)$ where τ_0 is the canonical bar twisting morphism.

2.5. **The PBW spectral sequence.** The PBW filtration on $\mathcal{A}$ induces a filtration on the bar complex $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ by bar degree, producing a spectral sequence

$$E_1^{p,q}(\mathcal{A}) \;=\; H^{p+q}(\mathrm{gr}^F_p\, \bar{B}^{\mathrm{ch}}(\mathcal{A})) \implies H^{p+q}(\bar{B}^{\mathrm{ch}}(\mathcal{A})).$$

On the E_1 page, $\mathrm{gr}^F_p\, \bar{B}^{\mathrm{ch}}(\mathcal{A})$ is computed from the Poisson vertex algebra $R_{\mathcal{A}}$ by the Chevalley–Eilenberg-style bar construction; the higher differentials d_r ($r \geq 2$) encode quantum corrections. Collapse at E_2 means $d_r = 0$ for all $r \geq 2$, which is condition (ii) of Theorem 3.1.

2.6. **The chiral Hochschild complex.** The chiral Hochschild complex $C^\bullet_{\mathrm{ch}}(\mathcal{A})$ has degree-n cochains

$$C^n_{\mathrm{ch}}(\mathcal{A}) \;=\; \Gamma\big(\overline{C}_{n+2}(X),\, j_*j^*\mathcal{A}^{\boxtimes(n+2)} \otimes \Omega^n_{\overline{C}_{n+2}(X)}(\log D)\big),$$

with differential $d_{\mathrm{Hoch}} = d_{\mathrm{int}} + d_{\mathrm{fact}} + d_{\mathrm{config}}$ (internal, factorization, configuration-space de Rham). Its cohomology $\mathrm{ChirHoch}^*(\mathcal{A})$ is the tangent complex to the moduli of chiral algebra structures at $\mathcal{A}$ [10].

3. The Koszulness Criteria

Theorem 3.1 (Equivalences and consequences of chiral Koszulness)**.** *Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve X with PBW filtration $F_\bullet$. For a smooth projective curve Σ, let $\mathsf{FH}_\Sigma(\mathcal{A})$ denote the filtered comparison between the geometric bar object and the factorization homology realization, with strong convergence for the bar-length and Goodwillie filtrations. Let $\mathsf{FH}^{\det}_\Sigma(\mathcal{A})$ denote the detecting strengthening: degree-zero concentration after realization reflects degree-zero concentration of the geometric bar model.*

Conditions (i)–(iii) *and* (v) *below are the intrinsic bidirectional equivalence core in the Theorem-B ambient. Condition* (iv) *is a consequence of the core; its converse requires the chiral BGS/Priddy diagonal-detection hypothesis. Condition* (vi) *is a consequence of* (v)*. Condition* (vii) *is a factorization-homology consequence under* FH*, and becomes a criterion only under* $\mathsf{FH}^{\det}$*. Condition* (viii) *is a one-way chiral Hochschild consequence on the Koszul locus. Condition* (ix) *is a bidirectional criterion on the families where a Kac–Shapovalov determinant controls the PBW comparison. Condition* (x) *is equivalent to the core when FM boundary detection holds. Under additional perfectness and non-degeneracy hypotheses on the ambient tangent complex, condition* (xi) *is also equivalent to the intrinsic core. Condition* (xii) *implies condition* (x)*; the converse is known for affine Kac–Moody at non-critical level and is open for Virasoro and $\mathcal{W}$-algebras.*

Intrinsic equivalence core and monadic consequence:

- (i) *$\mathcal{A}$ is chirally Koszul (Definition 2.3).*
- (ii) *The PBW spectral sequence on $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ collapses at E_2.*
- (iii) *The minimal A_∞-model of $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ is formal: $m_n = 0$ for $n \geq 3$.*
- (iv) *Ext diagonal vanishing: $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X, \omega_X) = 0$ for $p \neq q$.*
- (v) *The bar–cobar counit $\Omega^{\mathrm{cont}}_X(C_{\mathcal{A}}) \to \mathcal{A}$ is a quasi-isomorphism, with $C_{\mathcal{A}} = \bar{B}^{\mathrm{ch}}(\mathcal{A})$ on raw finite PBW windows and $C_{\mathcal{A}} = \widehat{\bar{B}^{\mathrm{ch}}}(\mathcal{A})$ on the strict completed pro-conilpotent regime.*
- (vi) *The Barr–Beck–Lurie comparison for the bar–cobar adjunction is an equivalence on the fiber over $\bar{B}^{\mathrm{ch}}(\mathcal{A})$.*
- (ix) *The Kac–Shapovalov determinant $\det G_h \neq 0$ in the bar-relevant range.*
- (x) *The restriction $i_S^! \bar{B}_n(\mathcal{A})$ to every FM boundary stratum S is acyclic outside degree* 0.

Factorization homology and Hochschild consequences:

- (vii) *If $\mathsf{FH}_\Sigma(\mathcal{A})$ holds, chiral Koszulness implies*

$$H^k\Big(\textstyle\int_\Sigma \mathcal{A}\Big) = 0 \qquad (k \neq 0).$$

  If $\mathsf{FH}^{\det}_{\mathbb{P}^1}(\mathcal{A})$ also holds, this genus-zero concentration is equivalent to the intrinsic core. Under the uniform-weight hypothesis, the all-genera scalar obstruction is $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A})\lambda_g$.
- (viii) *On the Koszul locus,* $\mathrm{ChirHoch}^\bullet(\mathcal{A})$ *has cohomological amplitude* $[0, 2]$*, Hochschild duality with $\mathcal{A}^!$, and Hilbert series determined by the degree* 0, 1, 2 *terms. A free-polynomial cup-product description requires the corresponding Massey-vanishing hypothesis.*

Conditional and partial:

(xi) Lagrangian criterion: *$\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ are transverse Lagrangians in the (-1)-shifted symplectic deformation space $\mathcal{M}_{\mathrm{comp}}$.*

(xii) $\mathcal{D}$-module purity: *each $\bar{B}^{\mathrm{ch}}_n(\mathcal{A})$ is pure of weight n as a mixed Hodge module, with characteristic variety aligned to FM boundary strata.*

4. Proof of the Koszulness criteria

The intrinsic circuit is

$$\text{(i)} \Leftrightarrow \text{(ii)} \Leftrightarrow \text{(iii)} \Leftrightarrow \text{(v)};$$

conditions (iv), (ix), and (x) are equivalent to nodes of this circuit. Conditions (vi), (vii), and (viii) are consequences with the hypotheses stated in Theorem 3.1. Condition (xi) is treated under the perfectness and non-degeneracy hypotheses. Condition (xii) is used in the proved direction toward boundary acyclicity.

4.1. **The core circuit.**

(i) $\Leftrightarrow$ (ii). Koszulness, in the sense of Definition 2.3, is by definition acyclicity of $K^L_\tau(\mathcal{A}, \bar{B}^{\mathrm{ch}}(\mathcal{A}))$ and $K^R_\tau(\bar{B}^{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ for the canonical twisting morphism τ_0. The left twisted tensor product is filtered by the PBW bar degree, and on the associated graded the twisted tensor product becomes the standard Koszul complex of the Poisson vertex algebra $R_{\mathcal{A}}$. Acyclicity on the associated graded lifts to acyclicity on K^L_τ if and only if the bar spectral sequence collapses at E_2 [10, Thm. 4.2.1]. This is the PBW Koszulness criterion, and it gives the first equivalence.

(ii) $\Rightarrow$ (v). E_2-collapse gives bar concentration on the diagonal, which is the input for bar–cobar inversion. Specifically, if the spectral sequence collapses, then the cohomology $H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ is concentrated in bidegree (p, p); the cobar functor on a diagonally concentrated bigraded coalgebra returns the algebra back, giving the continuous counit $\Omega^{\mathrm{cont}}_X(C_{\mathcal{A}}) \xrightarrow{\sim} \mathcal{A}$ in the Theorem-B ambient. Here $C_{\mathcal{A}} = \bar{B}^{\mathrm{ch}}(\mathcal{A})$ on raw finite PBW windows and $C_{\mathcal{A}} = \widehat{\bar{B}^{\mathrm{ch}}}(\mathcal{A})$ on the strict completed pro-conilpotent regime.

(v) $\Rightarrow$ (i). If the counit is a quasi-isomorphism, then its mapping cone is acyclic. The mapping cone of $\varepsilon_{\mathcal{A}}$ is canonically quasi-isomorphic to the twisted tensor product K^L_τ (a standard lemma about bar–cobar; see [11, §2.2] in the associative case, adapted to the chiral setting in [10]). Acyclicity of K^L_τ is the left twisted tensor part of Definition 2.3; the right twisted tensor product follows by the same universal twisting-morphism theorem and the PBW convergence hypothesis.

(v) $\Rightarrow$ (viii). The quasi-isomorphism $\Omega^{\mathrm{ch}}(\bar{B}^{\mathrm{ch}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ is a free resolution of $\mathcal{A}$ by a chiral cofree algebra. Computing chiral Hochschild cohomology through this resolution turns $\mathrm{ChirHoch}^*(\mathcal{A})$ into a bigraded complex whose E_1-page is the cotangent complex of the bar coalgebra. The bar-cobar filtration gives cohomological amplitude $[0, 2]$ and the Hochschild duality pairing with the Verdier dual Koszul algebra $\mathcal{A}^!$. A free-polynomial product statement is stronger; it follows only under the Massey-vanishing hypothesis for the transferred Fulton–MacPherson product. Chiral Hochschild concentration alone does not imply the counit criterion here.

(ii) $\Leftrightarrow$ (iii). For the forward direction, E_2-collapse of the PBW spectral sequence means all differentials $d_r = 0$ for $r \geq 2$. The Homological Perturbation Lemma identifies the differentials d_r of the spectral sequence with the A_∞ operations m_{r+1} of the minimal model under the transfer along the PBW filtration, up to sign. Hence $d_r = 0$ for $r \geq 2$ is equivalent to $m_n = 0$ for $n \geq 3$.

For the reverse direction, if the minimal A_∞-model is formal then $H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ is a strict dg algebra with no higher operations, and its bar spectral sequence collapses at E_2 by Keller's theorem on formal A_∞-algebras [8]. The argument proceeds fiberwise on FM strata; each fiber is an A_∞-algebra over a field, to which Keller's theorem applies directly. The PBW filtration is compatible with the FM stratification, so fiberwise collapse assembles to global collapse.

4.2. Ext diagonal vanishing and the core.

(i) $\Rightarrow$ (iv). E_2-collapse places $H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ on the diagonal $p = q$; the bar complex computes $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X, \omega_X)$, so $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X, \omega_X) = 0$ for $p \neq q$.

(iv) $\Rightarrow$ (i). Assume the chiral BGS/Priddy diagonal-detection hypothesis for the chosen finite PBW component. If the Ext bigrading is diagonal, then $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ has cohomology concentrated on the line $p = q$. A d_r-differential in the PBW spectral sequence moves off-diagonal by $(r, 1 - r)$, which would produce off-diagonal classes, contradicting diagonal concentration of the abutment. Hence $d_r = 0$ for all $r \geq 2$.

The spectral sequence converges conditionally in the sense of Boardman (filtration Hausdorff and complete, since the PBW filtration on a positively graded chiral algebra is bounded below). In characteristic zero with a split filtration (the PBW filtration splits by conformal weight), the extension problem is trivial: $E_\infty = E_2$ as bigraded objects. So E_2-collapse gives (ii), which is equivalent to (i).

4.3. Factorization homology concentration.

(i) $\Rightarrow$ (vii). Assume $\mathsf{FH}_\Sigma(\mathcal{A})$. The filtered comparison identifies the associated graded of the Goodwillie filtration on $\int_\Sigma \mathcal{A}$ with the corresponding geometric bar contributions. PBW collapse concentrates those contributions in degree 0, hence $H^k(\int_\Sigma \mathcal{A}) = 0$ for $k \neq 0$. Under the uniform-weight hypothesis, the surviving scalar obstruction is $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A})\lambda_g$.

(vii) $\Rightarrow$ (i) *under detection.* Assume in addition $\mathsf{FH}^{\det}_{\mathbb{P}^1}(\mathcal{A})$. Then degree-zero concentration of $\int_{\mathbb{P}^1} \mathcal{A}$ reflects degree-zero concentration of $\bar{B}^{\mathrm{ch}}_{\mathbb{P}^1}(\mathcal{A})$. This is PBW E_2 collapse, hence condition (ii) and therefore chiral Koszulness.

4.4. FM boundary acyclicity.

(i) $\Rightarrow$ (x). Each FM boundary stratum $S_T \cong \prod_{v \in V(T)} \overline{C}_{|v|}(X_v)$ is indexed by a rooted tree T. The residue component $d_{\mathrm{res}}|_{S_T}$ of the bar differential restricts to the tensor product of lower-degree bar complexes, one for each vertex of T. E_2-collapse at each vertex degree gives acyclicity at S_T: $H^k(i^!_{S_T} \bar{B}_n(\mathcal{A})) = 0$ for $k \neq 0$.

(x) $\Rightarrow$ (i). At the deepest stratum, the binary collision divisor $S = D_{\{1,2\}} \cong X \times \overline{C}_{n-1}(X)$, the restriction $i^!_S \bar{B}_n(\mathcal{A})$ computes the bar-complex contribution from the OPE $a_{(k)}b$ at a single collision point. Acyclicity of $i^!_S$ at every stratum forces every residue to be exact, which is the PBW condition $d_1(e_i) = 0$ on the associated graded. Iterating over all strata gives E_2-collapse, hence (i).

4.5. Barr–Beck–Lurie monadicity.

(v) ⇒ (vi). The Barr–Beck–Lurie theorem [6, Thm. 4.7.3.5] states that the comparison functor Φ associated with the bar–cobar adjunction is an equivalence if and only if the bar functor $\bar{B}_\kappa$ is conservative and preserves totalizations of $\bar{B}_\kappa$-split cosimplicial objects.

Conservativity holds on the Koszul locus: $\bar{B}_\kappa$ detects quasi-isomorphisms, because the bar–cobar counit inverts them; this is condition (v).

Totalization preservation: at genus 0 this follows from the bar–cobar counit. At all genera it is conditional on the Francis–Gaitsgory $(\infty, 2)$ descent package for the completed ambient.

Therefore the BBL comparison is an equivalence whenever the counit criterion holds and the required totalization preservation is available. The reverse implication is not used as an independent criterion.

4.6. Kac–Shapovalov determinant.

(i) ⇒ (ix). If $\mathcal{A}$ is chirally Koszul, the PBW filtration is strict on the vacuum module: $\mathrm{gr}^F M_0(\mathcal{A}) \cong \mathrm{Sym}^{\mathrm{ch}}(V)$ where $V = F_1\mathcal{A}/F_0\mathcal{A}$. Strictness of the PBW map implies that the Shapovalov form G_h is non-degenerate in the bar-relevant range.

(ix) ⇒ (i). Conversely, non-degeneracy of G_h in the bar-relevant range means the PBW-to-bar comparison map $\iota\colon \mathrm{Sym}^{\mathrm{ch}}(V)_h \to \bar{B}_h(\mathcal{A})$ is injective at every bar-relevant weight h. Suppose for contradiction $d_r \neq 0$ for some $r \geq 2$. Then there exists $x \in E_r^{p,q}$ with $d_r(x) \neq 0$; let $\tilde{x} \in F^p\bar{B}$ be a lift. If $\tilde{x} \in \mathrm{im}(\iota)$, it represents a nonzero class in $\mathrm{gr}^p(\bar{B}) \cong \mathrm{Sym}^p(V)$, and $d_r(\tilde{x})$ represents a nonzero class in gr^{p+r}. By PBW strictness (injectivity of ι), this contradicts the acyclicity of $\mathrm{Sym}(V)$ as a Koszul complex. Hence $d_r = 0$ for all $r \geq 2$, giving (ii), hence (i).

4.7. The Lagrangian criterion, conditionally.

(i) ⇒ (xi). Under the perfectness and non-degeneracy hypotheses on the ambient tangent complex [10, Prop. 6.3]: on the Koszul locus, the bar-cobar adjunction provides a free resolution of $\mathcal{A}$, and the complementarity splitting

$$\mathbf{Q}_g(\mathcal{A}) \oplus \mathbf{Q}_g(\mathcal{A}^!) \;\simeq\; H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$$

identifies $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ as complementary subspaces of $\mathcal{M}_{\mathrm{comp}}$. The (-1)-shifted symplectic structure on $\mathcal{M}_{\mathrm{comp}}$ makes each of these subspaces isotropic; complementarity gives maximal dimension, hence both are Lagrangian.

(xi) ⇒ (i). If $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ are transverse Lagrangians and the perfect comparison identifies their normal intersection complex with the Koszul cone, then their derived intersection

$$\mathcal{M}_{\mathcal{A}} \times^h_{\mathcal{M}_{\mathrm{comp}}} \mathcal{M}_{\mathcal{A}^!}$$

is discrete: transverse Lagrangian intersection in a (-1)-shifted symplectic space has expected dimension 0. The normal comparison identifies discreteness of this intersection with acyclicity of the chiral Koszul cone for the twisting datum $(\mathcal{A}, \bar{B}^{\mathrm{ch}}(\mathcal{A}), \tau_0)$. That acyclicity is Definition 2.3.

4.8. $\mathcal{D}$-module purity, one direction.

(xii) ⇒ (x). Suppose each $\bar{B}_n^{\mathrm{ch}}(\mathcal{A})$ is pure of weight n as a mixed Hodge module. By Saito's theory [14], the weight filtration on a pure Hodge module splits up to semisimplification; for the bar complex, this splitting implies that the restriction $i_S^!\,\bar{B}_n(\mathcal{A})$ to any FM boundary stratum S is again pure. Purity plus the alignment of the characteristic variety to FM strata forces acyclicity outside degree 0 on every stratum, which is condition (x).

Remark 4.1 (Converse direction). The converse, that Koszulness implies $\mathcal{D}$-module purity of the bar complex, is established for the affine Kac–Moody family in [10, Prop. 7.5]: there the bar complex is identified with a geometric complex whose purity is known via the Riemann hypothesis for weight filtrations. For Virasoro and the $\mathcal{W}$ algebras, the required geometric origin for the bar complex is not known, and the converse is open.

5. Examples

The following standard families give the first tests of the intrinsic core.

5.1. Heisenberg. The Heisenberg vertex algebra Heis_k at level k is generated by a single current $J(z)$ with OPE

$$J(z)J(w) \;\sim\; \frac{k}{(z-w)^2}.$$

At nonzero level, the PBW filtration is strict: the associated graded is a free polynomial Poisson vertex algebra. The Kac–Shapovalov determinant is a product of factors $(n-h)^{\mu_n}$ for the weight lattice, and is nonzero generically. Condition (ix) holds, and Theorem 3.1 supplies the other core conditions on this row.

In the shadow tower language, Heis_k is a Gaussian algebra (depth $r_{\max} = 2$): the bar complex is abelian and the chiral Hochschild cohomology is concentrated in the single bar degree $\{0, 1, 2\}$ that is visible on any free theory. The A_∞-structure is strictly associative, and the bar–cobar counit is a quasi-isomorphism by direct computation.

Remark 5.1. The Heisenberg collision residue is $r^{\mathrm{coll}}_{\mathrm{Heis}}(z) = k/z$. The shadow constant is $\kappa^{\mathrm{Heis}} = k$. At $k = 0$ the double-pole residue and κ vanish; the reduced bar remains the Gaussian free-field bar complex.

5.2. Affine Kac–Moody at generic level. Let $\hat{\mathfrak{g}}_k$ be the affine Kac–Moody vertex algebra of a simple Lie algebra $\mathfrak{g}$ at level k. The Kac–Shapovalov determinant is given by the Kac determinant formula; it vanishes precisely at the Shapovalov walls $k + h^\vee = p/q$ for a discrete set of rationals depending on the weight. Away from these walls, condition (ix) holds, and Theorem 3.1 gives chiral Koszulness.

For the bar–cobar pair $(\hat{\mathfrak{g}}_k, \hat{\mathfrak{g}}_{k'})$ with $k + k' = -2h^\vee$ (the Killing self-duality), the shadow constant κ is

$$\kappa^{\mathrm{KM}}(\mathfrak{g}, k) \;=\; \frac{\dim(\mathfrak{g})\,(k + h^\vee)}{2h^\vee}.$$

At the critical level $k = -h^\vee$, the shadow constant κ^{KM} vanishes; the Shapovalov determinant acquires zeros of the corresponding order, and the Kac–Shapovalov input (condition (ix)) enters its degenerate regime. The critical level lies outside the generic Koszul locus treated here. At generic $k \neq -h^\vee$, $\kappa^{\mathrm{KM}} \neq 0$ and the PBW criterion applies.

For $\mathfrak{g} = \mathfrak{sl}_2$ and $k = 1$,

$$\kappa^{\mathrm{KM}} = \frac{3(1+2)}{2 \cdot 2} = \frac{9}{4}.$$

This agrees with the direct computation of the second shadow coefficient from the OPE [10, §3.1], and at $k = -h^\vee = -2$ one recovers $\kappa^{\mathrm{KM}} = 0$, reproducing the critical-level degeneration. At level $k = 0$ the trace-form double-pole collision residue $k\Omega_{\mathrm{tr}}/z$ vanishes, but the simple-pole Lie bracket channel remains. Thus $V_0(\mathfrak{g})$ is not Gaussian or abelian, and $\kappa(V_0(\mathfrak{g})) = \dim(\mathfrak{g})/2$.

5.3. Virasoro in the Koszul locus. The Virasoro vertex algebra Vir_c at central charge c has shadow constant $\kappa^{\mathrm{Vir}} = c/2$. The Kac determinant formula for Vir_c gives the discrete set of degenerate central charges $c_{p,q} = 1 - 6(p-q)^2/(pq)$; away from these, the vacuum module is irreducible in the bar-relevant range, and condition (ix) holds.

At $c = 13$, the Koszul pair $(\mathrm{Vir}_c, \mathrm{Vir}_{26-c})$ is self-dual, and the duality constant $c + c' = 26$ (the Koszul conductor of Virasoro) matches the matter plus bc ghost anomaly-cancellation normalization. At these generic values, Vir_c is chirally Koszul and the PBW/Kac criterion applies.

Virasoro has one strong generator. Hence $\delta F_g^{\mathrm{cross}}(\mathrm{Vir}_c) = 0$ for every g, while the class-M shadow tower remains infinite through the nonzero higher Virasoro shadows.

5.4. **The $\beta\gamma$ system.** The $\beta\gamma$ system is a free-field vertex algebra with commuting fields $\beta(z), \gamma(z)$ satisfying $\beta(z)\gamma(w) \sim 1/(z-w)$. Its associated graded is the Poisson vertex algebra on $T^*(\mathcal{O}_X)$ with the canonical bracket; the bar complex is an exterior algebra on cotangent generators.

The PBW bar model of the free $\beta\gamma$ system is formal: the transferred higher A_∞ operations vanish on the free bar generators. This proves condition (iii). The shadow-tower statement is separate: $\beta\gamma$ is contact class C, with depth $r_{\max} = 4$ and a nonzero quartic contact shadow.

5.5. **Summary.**

Family	Decisive criterion	Conclusion
Heisenberg	free PBW bar	Gaussian Koszul locus
Affine Kac–Moody	Kac–Shapovalov nondegeneracy	generic Koszul locus
Virasoro	PBW collapse off Kac walls	class-M Koszul locus
$\beta\gamma$ system	free A_∞ bar formality	contact Koszul locus

For these families, one decisive criterion enters the intrinsic core. The Lagrangian criterion (xi) is available where the perfectness and non-degeneracy hypotheses are verified. $\mathcal{D}$-module purity (xii) is proved for affine Kac–Moody at non-critical level and remains open for Virasoro and $\mathcal{W}$.

6. Open questions

6.1. **The converse of $\mathcal{D}$-module purity.** The remaining open equivalence is (xii). The implication (xii) $\Rightarrow$ (x) is unconditional (Section 4); the converse is known for the affine Kac–Moody family (Remark 4.1) but open for Virasoro and $\mathcal{W}$.

Conjecture 6.1. *For every chirally Koszul chiral algebra $\mathcal{A}$ on a smooth projective curve, each $\bar{B}_n^{\mathrm{ch}}(\mathcal{A})$ is pure of weight n as a mixed Hodge module, with characteristic variety aligned to the FM boundary strata.*

The obstacle is the absence of a known geometric model from which purity of the Virasoro bar complex can be read off. Any proof of the converse must proceed either by a direct computation of the weight filtration on $\bar{B}_n^{\mathrm{ch}}(\mathrm{Vir}_c)$, or by an indirect route via a geometric model (e.g., a resolution of the Virasoro bar complex by a pure complex on a related moduli space).

6.2. **The perfectness hypothesis for (xi).** Condition (xi) requires perfectness and non-degeneracy of the ambient tangent complex, which is verified for the standard landscape [10, Prop. 6.3] but not known in full generality. A natural question is whether perfectness holds for every chirally Koszul algebra.

Conjecture 6.2. *If $\mathcal{A}$ is chirally Koszul, the ambient tangent complex at $\mathcal{A}$ in $\mathcal{M}_{\mathrm{comp}}$ is perfect of tor-amplitude* $[0, 2]$.

If this conjecture holds, the Lagrangian criterion (xi) is equivalent to the intrinsic Koszulness core.

6.3. **The Koszul locus beyond the standard landscape.** A natural family of chiral algebras outside the standard landscape consists of the non-principal $\mathcal{W}$-algebras of type ADE. The Koszul duality for hook-type $\mathcal{W}$-algebras of type A has been established in [10]; other nilpotent orbits remain open and are controlled by the associated variety hierarchy recorded in [10].

6.4. **Shadow depth and Koszulness.** The shadow depth classification (Gaussian, Lie, contact, mixed) and chiral Koszulness are logically distinct: Koszulness concerns the diagonal concentration of the bar cohomology, while shadow depth concerns the degree at which the shadow tower terminates. The standard generic rows are chirally Koszul. Gaussian (Heisenberg) and contact ($\beta\gamma$) rows have finite shadow depth; Virasoro and $\mathcal{W}$-algebras have $r_{\max} = \infty$ (class M). The Koszulness criteria do not distinguish these shadow classes.

Remark 6.3 (Koszulness is not SC formality). Chiral Koszulness is an assertion about H^* in diagonal bidegree. Swiss-cheese formality is the vanishing of higher operations on the open-closed bar complex. The standard generic families lie on the chiral Koszul locus; Swiss-cheese formality is restricted to class G.

Appendix A. Scope of the Auxiliary Criteria

At genus 0, PBW E_2 collapse is the common hub for the Koszul twisting datum, the bar–cobar counit, twisted tensor acyclicity, Priddy bar-weight concentration, and the Hochschild amplitude consequence. Swiss-cheese formality coincides with this hub only on class G; classes L, C, and M may be Koszul and fail Swiss-cheese formality.

The Barr–Beck–Lurie comparison is used as a monadic consequence of the bar–cobar counit. Its all-genera form inherits the Francis–Gaitsgory $(\infty, 2)$ descent hypotheses of the completed ambient. Factorization homology is a criterion only when the filtered comparison FH and its detecting strengthening $\mathsf{FH}^{\mathrm{det}}$ are available. Under the uniform-weight hypothesis the surviving scalar class is $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A})\lambda_g$; multi-weight families such as $\mathcal{W}_N$ require the cross-channel correction beginning with $\delta F_g^{\mathrm{cross}}$.

The Hochschild condition is one-way. PBW Koszulness gives cohomological amplitude $[0, 2]$ and duality for $\mathrm{ChirHoch}^\bullet(\mathcal{A})$ on the Koszul locus. A free-polynomial cup-product algebra is stronger and requires Massey vanishing for the transferred Fulton–MacPherson product.

The Lagrangian criterion is equivalent to the intrinsic core only when the ambient tangent complex is perfect and non-degenerate and the normal-intersection comparison identifies the derived intersection with the chiral Koszul cone. The $\mathcal{D}$-module purity criterion is proved in the direction (xii)$\Rightarrow$(x); the converse is known for affine Kac–Moody at non-critical level and remains open in the Virasoro and $\mathcal{W}$ cases.

For admissible levels $k + h^\vee = p/q$, the periodic-CDG comparison hypothesis supplies a Koszul-compatible filtration on the non-degenerate simply-laced admissible case at the level of cohomology of filtration quotients. Strict chain-level periodicity still depends on Massey vanishing, and descent from the small-quantum-group stabilization to the semisimplified Kazhdan–Lusztig target remains a separate conjectural step.

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